

ering

Schedule



zain@Zain-Laptop: ~/xv6-riscv/xv6-labs-2026

s fork, exec, and wa

find, which execut

-exec behavior:

hi" and the file is "

ommand on each file.w

, which may be useful

shell script findtest.

5
100
18
6
\$ memdump
Example 1:
61810
2026
Example 2:
a string
Example 3:
another
Example 4:
0x00000000000000C80
1819438967
100
z
xyzzzy
Example 5:
hello
w
o
r
l
d
\$ find . wc -exec echo hi
hi ./wc
\$

31771

zain aleem

u), it will build a new

ave a look at K&R (t

re strings as in Pyth

Makefile.

following output (w

```
zombie      2 24 34848
logstress   2 25 37560
forphan     2 26 36352
tdorphan    2 27 35800
sync        2 28 34944
memdump     2 29 44072
console     3 30 0
```

```
$ find grep
usage: find path name [-exec cmd args...]
$ find .grep
usage: find path name [-exec cmd args...]
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
$ find .b
usage: find path name [-exec cmd args...]
$ find . b
./zombie
./b
./a/b
./a/aa/b
$
```

31771

zain aleem

```
egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,dr
```

xv6 kernel is booting

hart 1 starting

hart 2 starting

init: starting sh

\$ sh < findtest.sh

hello

hello

hello

\$ echo DONE

DONE

\$ uptime

uptime: 963 ticks

\$ find . uptime

./uptime

\$ find . '^uptime\$'

\$ find . '.ptime'

\$ find . ^uptime\$

./uptime

\$ find . ^_uptime\$

\$ find . ^u.time\$

./uptime

\$ find . ^up*time\$

./uptime

\$ find . xyz

\$ find . time

./uptime

31771

zain aleem

```
303 while(getcmd(buf, sizeof(buf)) >= 0){
```

```
310 if
```

```
314
```

```
315
```

```
316
```

```
317 if
```

```
318
```

```
319
```

```
320
```

```
321
```

```
322
```

```
323
```

```
324
```

```
325
```

```
326
```

```
327
```

```
328
```

```
329
```

```
330
```

```
331
```

```
332
```

```
333
```

```
334
```

```
335
```

```
336
```

```
void
```

```
kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-leg
```

```
acy=false -drive file=fs.img,if=none,format=raw,id=x0 -device
```

```
virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
```

```
xv6 kernel is booting
```

```
hart 1 starting
```

```
hart 2 starting
```

```
init: starting sh
```

```
$ echo one
```

```
one
```

```
$ echo two
```

```
two
```

```
$ echo three
```

```
three
```

```
$ history
```

```
1 echo one
```

```
2 echo two
```

```
3 echo three
```

```
$ !!
```

```
echo three
```

```
three
```

```
$ !1
```

```
echo one
```

```
one
```

```
$ !99
```

```
sh: no such command in history
```

zain@Zain-Laptop: ~/xv6-riscv/xv6-labs-2026

31771

zain aleem

no arguments

```
5
100
18
6
$ memdump
Example 1:
61810
2026
Example 2:
a string
Example 3:
another
Example 4:
0x00000000000000C80
1819438967
100
z
xyzzzy
Example 5:
hello
w
o
r
l
d
$
```

31771

zain aleem

```

32 prefixlen = ta
33 uptime
34 find
35 if((fd = open(cat
36     fprintf(2, "sixfive
37     // just drop sleep
38     memmove(tab, echo
39     return 1; forktest
40 } grep
41 init
42 kill
43 while(read(fd, ln
44     if(de.inum = ls
45         continue; mkdir
46     if(strlen(de rm
47         continue; sh
48     stressfs
49 if(prefix_ma usertests
50     if(nmatche grind
51         memmove(wc
52         matches[ zombie
53         nmatches logstress
54     } forphan
55     } dorphan
56     } sync
57     } memdump
58     } console

```

31771

zain aleem

```

2 5 35088
2 6 48680
2 7 36728
2 8 38432
2 9 35368
2 10 35592
2 11 18136
2 12 44184
2 13 35976
2 14 35512
2 15 35304
2 16 42880
2 17 35568
2 18 35552
2 19 62416
2 20 36416
2 21 205880
2 22 51824
2 23 37728
2 24 34848
2 25 37560
2 26 36352
2 27 35800
2 28 34944
2 29 44072
3 30 0

```

ring

Sche



zain@Zain-Laptop: ~/xv6-riscv/xv6-labs-2026

five must pri
characters in the
and-line argum
ch open fails,

ve's behavior:

ne time
any of the sep
parators.

```
zain@Zain-Laptop:~/xv6-riscv/xv6-labs-2026$ make qemu
```

```
mv -f fs.img fs.img.bk
```

```
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_uptime  
user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/  
user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/  
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1
```

```
ballocc: first 1191 blocks have been allocated
```

```
ballocc: write bitmap block at sector 46
```

```
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m  
legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio
```

```
xv6 kernel is booting
```

```
hart 2 starting
```

```
hart 1 starting
```

```
init: starting sh
```

```
$ sleep 10
```

```
$ sixfive sixfive.txt
```

```
5
```

```
100
```

```
18
```

```
6
```

```
$
```

31771

zain aleem

ng

Sche



zain@Zain-Laptop: ~/xv6-riscv/xv6-labs-2026

1 of the xv6

t some of th

Makefile; onc

nt, sleep sho

d as a string

nel code that

e assembler

The C progra

zain@Zain-Laptop:~/xv6-riscv/xv6-labs-2026\$ make qemu

mv -f fs.img fs.img.bk

mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_upt

er/_echo user/_forktest user/_grep user/_init user/_kill user/_ln us

user/_usertests user/_grind user/_wc user/_zombie user/_logstress us

meta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks

ballocc: first 1191 blocks have been allocated

ballocc: write bitmap block at sector 46

qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel

legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virt

xv6 kernel is booting

hart 2 starting

hart 1 starting

init: starting sh

\$ sleep 10

\$

31771

zain aleem

while)

zain@Zain-Laptop: ~/xv6-riscv/xv6-labs-2026

zain@Zain-Laptop:~/xv6-riscv/xv6-labs-2026\$ make qemu

mv -f fs.img fs.img.bk

mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_uptime user/_find user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkd user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 t

ballocc: first 1191 blocks have been allocated

ballocc: write bitmap block at sector 46

qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -no legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,dr

xv6 kernel is booting

hart 1 starting

hart 2 starting

31771

init: starting sh

zain aleem

\$ sh < findtest.sh

hello

hello

hello

\$ echo DONE

DONE

\$ uptime

uptime: 963 ticks

\$

 zain@Zain-Laptop: ~/xv6-riscv/xv6-labs-2026

```
-fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-strmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/sh.o user/sh.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_sh user/sh.o user/ser/umalloc.o
riscv64-linux-gnu-objdump -S user/_sh > user/sh.asm
riscv64-linux-gnu-objdump -t user/_sh | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > u
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_uptime user/_find use
er/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkd
user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 t
ballocc: first 1191 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -no
egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,dr
```

xv6 kernel is booting

```
hart 1 starting
hart 2 starting
init: starting sh
$ sleep 100 &
$ wait
$
```

31771

zain aleem



zain@Zain-Laptop: ~/xv6-riscv/xv6-labs-2026

h for this lab

kernel/entry.

or -O -fno-

size=4096 -

/_zombie >

/_zombie |

test.sh use

0 inode blo

allocated

r 45

bios none -

zain@Zain-Laptop:~\$ cd xv6-riscv

zain@Zain-Laptop:~/xv6-riscv\$ ls

LICENSE Makefile README fs.img kernel mkfs test-xv6.py user x

zain@Zain-Laptop:~/xv6-riscv\$ xv6-labs-2026

xv6-labs-2026: command not found

zain@Zain-Laptop:~/xv6-riscv\$ cd xv6-labs-2026

zain@Zain-Laptop:~/xv6-riscv/xv6-labs-2026\$ make qemu

mv -f fs.img fs.img.bk

mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_uptin

er/_echo user/_forktest user/_grep user/_init user/_kill user/_ln use

user/_usertests user/_grind user/_wc user/_zombie user/_logstress use

nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks

ballocc: first 1191 blocks have been allocated

ballocc: write bitmap block at sector 46

qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m

egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virti

xv6 kernel is booting

hart 1 starting

hart 2 starting

init: starting sh

\$ ls

. 1 1 1024

.. 1 1 1024

README 2 2 2441

31771

zain aleem